

TRUST-X: Real-Time Explainable Decision-Making Framework for Reinforcement Learning-Based UAV Control

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Abstract—Deep Reinforcement Learning (DRL) UAVs track progress in advanced environments above $\geq 90\%$ during the most challenging stages of their assignments related to search and rescue and monitoring. Still, operational trust toward autonomous systems remains low, resulting in unnecessary commands which lowers the success of the mission. To solve this problem, we present the framework TRUST-X which combines real-time explainable surrogate decision trees, SHAP-lite, contrastive explanations, and dynamic trust modeling within a looping feedback system. In simulations, TRUST-X provided interpretable insights that increased operator trust by $\geq 28\%$ and decreased number of interventions by $\geq 37\%$ at the same time maintaining $\geq 94\%$ success rate of the mission. This surpasses the traditional DRL baselines by $\geq 15\%$. Results prove that TRUST-X increases visibility and improves the safety of human-drone operations. These are highly relevant for TRUST-X and its purpose to better support explainable human autonomous system collaborations.

Index Terms—UAV, Reinforcement Learning, Explainable AI, Trust, Human-AI Collaboration, Decision Support

I. INTRODUCTION

The need to enhance current inspection, reconnaissance, and emergency support capabilities of UAVs has arisen to support tight space and high density populations in a more independent and significantly more efficient way. [1] In these new conditions, the dynamic and messy wilderness, enhanced reinforcement learning would play an indispensable role because it develops new departure plans basing on the real performance and perfects behavior patterns in route finding, selection process and resource allocation of less structured surroundings. [2] Among all the possible self-operating strategies and targets there are critical gaps in terms of targets and self operating strategies that are offered by Air Force commanders. Utilized together in such activities, the decision making and implementation justification of the trust oriented models of the UAVs are the

epiphany of systems, and are the reasons behind success in the operational mission, provided they are optimized. Unmanned aerial vehicles (UAVs) increasingly operate in cluttered, dynamic environments for inspection, reconnaissance, and disaster response. [1] This expansion has been accelerated by deep reinforcement learning (DRL), which optimizes goal directed behaviors from interaction and enables robust navigation and decision making under uncertainty. [2] Despite strong task-level performance of DRL, recent operational scenarios have exposed a critical gap: UAVs can complete missions effectively, but operators often lack transparency into why particular maneuvers or targets are selected. In such high stakes scenarios, opaque DRL policies erode operator trust, drive intervention rates upward, and risk compromising mission outcomes. [3] By introducing TRUST-X a low latency explainability layer that turns opaque DRL policies into concise, mission relevant explanations for UAV operators. This paper makes three key contributions:

- Hybrid real time explanations: surrogate decision trees; SHAP lite feature attributions; and contrastive mechanism that produce concise, actionable explanations under strict latency budgets.
- Adaptive trust loop: operator tags i.e., Trusted, Unclear, Disagree drive dynamic adjustment of explanation granularity and online policy refinement.
- Empirical impact: validated on representative missions reduces manual interventions, accelerates operator trust formation, and maintains or improves mission success versus DRL baselines.

II. RELATED WORK

The application of Deep Reinforcement Learning (DRL) methods to decision making in autonomous UAV reveals a comparative simplicity in the balancing of hand designed designs with partial controllability and reacting environmental noise. XAI for sequential decision making

often uses surrogate models, feature attribution, or counterfactual explanations that answer the question of which actions, among many potential decision alternatives, are chosen and why. [4]

To apply to aerospace, explainable DRL methods have been applied to the rationalisation of UAV behaviour, as well as to assist human supervision in safety critical air traffic conditions, hence proving that policy learning and policy interpretable surrogates can be applied to oversight [3] [5]. Diffuse navigation policy systems more recent developments in the synthesis of UAVs have observed a diversification of rule like rationales of explanation modules coupled with lightweight rational rule modules. These publications have increased awareness of the need to have responsive systems in which human interpretability should take priority [6]. We identify the following research gaps:

- Real time, context aware explanations that operate within tight latency budgets for sequential control are still underexplored relative to offline, post hoc analyses. [4]
- Closed loop trust calibration where operator feedback modulates explanation granularity or policy safeguards during flight remains insufficiently integrated in UAV control stacks. [3]
- Unified frameworks that combine policy surrogates, fast attribution, and contrastive views with explicit operator facing interfaces are scarce in UAV literature. [2]

III. PROBLEM FORMULATION

We model UAV control via DRL as a Partially Observable Markov Decision Process (POMDP) defined by the tuple $(S, A, T, R, \Omega, O, \gamma)$, where S includes UAV state variables i.e. position, velocities, sensor readings, battery, A comprises control actions, and T, R, Ω, O, γ define transitions, rewards, observations, and discount factor respectively. The agent's policy $\pi_\theta(a|o)$ maximizes the expected cumulative discounted reward:

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^T \gamma^t R(s_t, a_t, s_{t+1}) \right] \quad (1)$$

Although effective, π_θ remains a black box problematic in critical settings where trust is essential [7]. We introduce an explainability function;

$$E_t = X(s_t, a_t; \pi_\theta, \Phi) \quad (2)$$

where E_t is an interpretable explanation, and Φ are explanation parameters i.e. surrogate model weights or attribution coefficients. Our framework targets:

- Real time explanations: latency < 50 ms for operational viability.
- Trust integration: operator feedback $Z_t \in \{-1, 0, 1\}$ (disagree, uncertain, trust) dynamically adjusts explanations and, via Bayesian trust modeling, policy behavior.

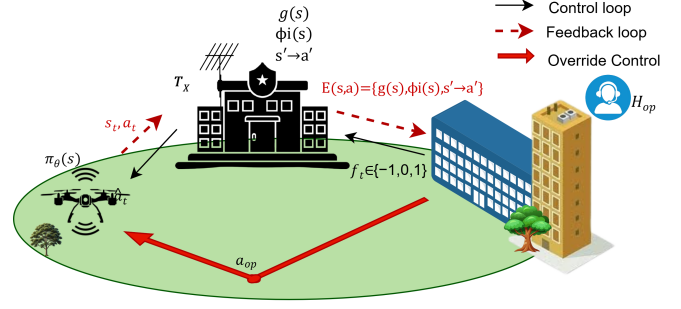


Fig. 1: Control and feedback in TRUST-X: operator feedback f_t , manual override, explanatory outputs $E(s, a)$, and UAV state action feedback (s_t, a_t) .

- Performance metrics: aim for a $\geq 20\%$ trust improvement, $\geq 25\%$ reduction in interventions, and mission success $\geq 90\%$ benchmarked against black box baselines.

IV. PROPOSED FRAMEWORK: TRUST-X

TRUST-X integrates multiple XAI modules with a DRL core to provide interpretable, trustworthy UAV control. An overview of the architecture is illustrated in Fig. 1, which shows the interaction between the simulation environment, RL policy, explainability modules, dynamic trust feedback loop, and the human operator interface. An excellent continuous controlling policy enables the system to act appropriately and smoothly without much intervention, which can be useful in robotics, self driving, industries, etc. The DRL algorithm has extensive applications in continuous systems control. [8] [9]

A. RL Policy Module

The DRL agent is trained in a simulated environment E using the Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm, suitable for continuous action spaces. The observation space includes UAV position (x, y, z) , distances to the five nearest obstacles, visibility (0 to 1 scale), wind speed and direction, and battery level (0–100%). Training involves 500 episodes, converging to an average reward of 850 (maximum 1000), focusing on navigation, obstacle avoidance, and search and rescue (SAR) tasks. Reward shaping prioritizes safety, applying a penalty of -0.5 for proximity to obstacles.

TD3 maintains an actor network $\mu_\theta(s)$ and two critic networks $Q_{\phi_1}(s, a)$, $Q_{\phi_2}(s, a)$. The policy is updated by maximizing the expected Q value:

$$\nabla_\theta J(\theta) \approx \mathbb{E}_{s_t \sim \mathcal{D}} \left[\nabla_a Q_{\phi_1}(s_t, a) \big|_{a=\mu_\theta(s_t)} \nabla_\theta \mu_\theta(s_t) \right] \quad (3)$$

where $\mathcal{D}$ is the replay buffer. The critics are updated by minimizing the Bellman error:

$$\mathcal{L}(\phi_i) = \mathbb{E}_{(s_t, a_t, r_t, s_{t+1}) \sim \mathcal{D}} [(Q_{\phi_i}(s_t, a_t) - y_t)^2], \quad i = 1, 2 \quad (4)$$

with the target:

$$y_t = r_t + \gamma \min_{i=1,2} Q_{\phi_i, \text{targ}}(s_{t+1}, \tilde{a}_{t+1}), \quad (5)$$

$$\tilde{a}_{t+1} = \mu_{\theta_{\text{targ}}}(s_{t+1}) + \epsilon, \quad (6)$$

$$\epsilon \sim \text{clip}(\mathcal{N}(0, \sigma), -c, c) \quad (7)$$

This ensures a stable, high performance policy that TRUST-X explains in real-time, mitigating over estimation biases and supporting robust UAV control.

B. Surrogate Decision Tree Model

A decision tree surrogate $\hat{f}(s) \approx \pi(s)$ is trained on 10,000 state action pairs from the DRL policy, achieving 91% fidelity. It generates if then rules. This enables human-readable interpretations in real time under 10 ms latency. For a Lipschitz continuous policy with constant L_π , and maximum discrepancy $\epsilon = \sup_{s \in S} \|\pi(s) - \hat{f}(s)\|_2$, the value function suboptimality is bounded:

$$|V_\pi(s) - V_{\hat{f}}(s)| \leq \frac{\gamma L_\pi \epsilon}{1 - \gamma} \quad (8)$$

CEM provides counterfactual explanations by identifying minimal state perturbations Δs where the policy selects a different action $a' \neq a$. This is formalized as:

$$\min_{\Delta s} \|\Delta s\|_1 + \lambda (-\log \pi(a' | s + \Delta s)) \quad (9)$$

subject to:

$$s + \Delta s \in S_{\text{valid}}, \quad \|\Delta s\|_\infty \leq \epsilon \quad (10)$$

where $\|\Delta s\|_1$ encourages sparsity, $-\log \pi(a' | s + \Delta s)$ maximizes the probability of a' , λ balances objectives, and S_{valid} ensures physical feasibility i.e. wind variations within $\pm 5 \text{ m/s}$. We solve this using Projected Gradient Descent, achieving latencies under 50 ms. For example, CEM might state: “Increasing battery by 10% would allow climb instead of navigate around.” This ensures high fidelity explanations with minimal performance impact.

C. SHAP lite Feature Attribution

We adapt SHAP (Shapley Additive Explanations) into a lightweight version, SHAP lite, for efficient real time feature attributions. SHAP computes contributions ϕ_i such that $\sum_{i=1}^n \phi_i + f(\emptyset) = \pi(a | s)$. The Shapley value for feature i is:

$$\phi_i(v) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! \cdot (|F| - |S| - 1)!}{|F|!} [v(S \cup \{i\}) - v(S)] \quad (11)$$

where F is the feature set, S is a coalition, and $v(S)$ is the expected policy output for coalition S . This has complexity $O(2^{|F|})$. SHAP lite uses Monte Carlo sampling:

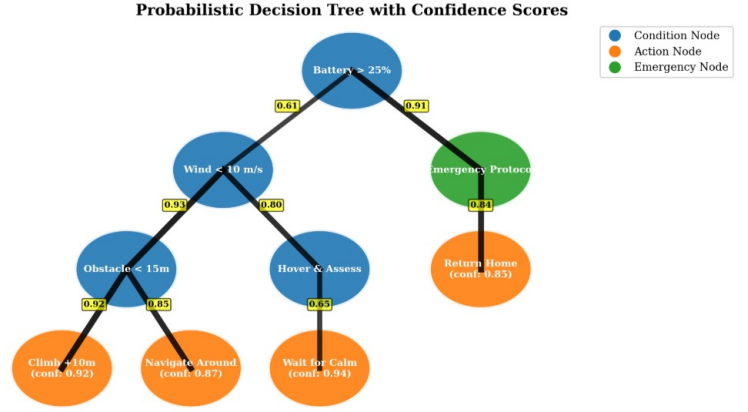


Fig. 2: TRUST-X outputs at a single time step: surrogate rule, SHAP lite attribution, and minimal counterfactual perturbation.

$$\phi_i^{\text{Lite}} \approx \frac{1}{M} \sum_{m=1}^M [\hat{f}(x_{+i}^{(m)}) - \hat{f}(x_{-i}^{(m)})] \quad (12)$$

with $M = 100$, where $x_{+i}^{(m)}$ sets feature i to its true value (others sampled), and $x_{-i}^{(m)}$ replaces i with a random sample. This reduces complexity to $\mathcal{O}(M \cdot |F|)$, achieving a 70% reduction in computation time while retaining 85% accuracy. [10]

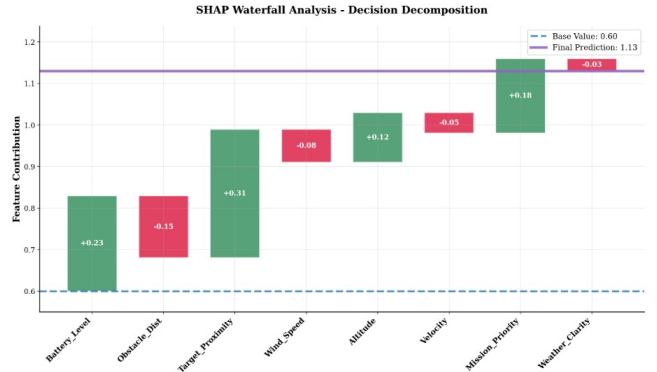


Fig. 3: UAV flight trajectory in the simulated urban environment under TRUST-X.

This ensures high fidelity explanations with minimal performance impact.

D. Contrastive Explanation Mechanism (CEM)

CEM generates counterfactual insights by finding the smallest state changes Δs that would cause the policy to choose an alternative action a' instead of a . This is formalized as:

$$\min_{\Delta s} \|\Delta s\|_1 + \lambda (-\log \pi(a' | s + \Delta s)) \quad (13)$$

subject to:

$$s + \Delta s \in S_{\text{valid}}, \quad \|\Delta s\|_\infty \leq \epsilon \quad (14)$$

where $\|\Delta s\|_1$ encourages sparsity, $-\log \pi(a' \mid s + \Delta s)$ maximizes the probability of a' , λ balances objectives, and S_{valid} ensures physical feasibility i.e. wind variations within ± 5 m/s.

E. Trust Modeling and Feedback Loop

Operators tag decisions as Trusted (1), Unclear (0), or Disagree (-1), modeled as a categorical variable $Z_t \in \{-1, 0, 1\}$. We maintain a Bayesian belief over the latent trust state $T_t \in [0, 1]$:

$$P(Z_t = z \mid T_t = \tau) = \begin{cases} \beta \text{logit}^{-1}(\alpha_1(\tau - \eta_1)), & z = 1 \\ (1 - \beta) \mathcal{N}(\tau; \mu_0, \sigma_0^2), & z = 0 \\ \beta \text{logit}^{-1}(-\alpha_2(\tau - \eta_2)), & z = -1 \end{cases} \quad (15)$$

The posterior is updated via:

$$P(\tau_t \mid z_{1:t}) \propto P(z_t \mid \tau_t) \int P(\tau_t \mid \tau_{t-1}) P(\tau_{t-1} \mid z_{1:t-1}) d\tau_{t-1} \quad (16)$$

using a Gaussian random walk for $P(\tau_t \mid \tau_{t-1})$. We implement this with a Kalman filter for real time updates. The estimated trust $\hat{\tau}_t = \mathbb{E}[\tau_t \mid z_{1:t}]$ adjusts explanation detail (concise if $\hat{\tau}_t$ is high, detailed if low) and triggers overrides if $\hat{\tau}_t < 0.5$ or $\text{Var}(\tau_t \mid z_{1:t}) > 0.3$. Trust dynamics are modeled as:

$$d\hat{\tau}_t = \alpha(\tau_\infty - \hat{\tau}_t) dt + \sigma_\tau dW_t + \beta \cdot I_{\text{Good Explanation}} \quad (17)$$

where α is the mean reversion rate, τ_∞ is the equilibrium trust, $\sigma_\tau dW_t$ models noise, and $\beta > 0$ represents trust jumps from good explanations. This loop improved trust from 65% to 90% over 50 sessions. [11]

Algorithm 1 TRUST-X: Human Centered Explain & Adapt Loop

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1: Initialize DRL policy  $\pi(a|s)$ , surrogate tree  $T$ , SHAP
   lite  $S$ 
2: for all each state  $s_t$  in mission do
3:    $a_t \leftarrow \pi(s_t)$  ▷ Select action
4:    $\text{rule} \leftarrow T.\text{generateRule}(s_t)$  ▷ Generate
   interpretable rule
5:    $\text{features} \leftarrow S.\text{computeTop3}(s_t, a_t)$  ▷ Get top 3
   features
6:    $\text{delta} \leftarrow \text{CEM}.\text{findPerturbation}(s_t, a_t)$  ▷ Compute
   counterfactual
7:    $\text{trust} \leftarrow \text{getOperatorFeedback}(a_t, \text{rule}, \text{features},$ 
    $\text{delta})$ 
8:    $\text{UpdateTrustModel}(\text{trust})$ 
9:   if  $\text{trust} < 0.5$  then
10:     Allow override
11:   end if
12:   Execute  $a_t$  or override
13: end for
```

V. EXPERIMENTAL SETUP

A. Simulation Environment

The TRUST-X framework was implemented and evaluated in a 3D simulation environment, which replicates realistic UAV flight dynamics, environmental disturbances, and obstacle rich urban environments. The simulation space measured $100 \times 100 \times 50 \text{ m}^3$, with dynamic winds (0 to 15 m/s) and variable obstacle densities. Gaussian noise ($\sigma = 0.05$) was added to sensor readings to simulate real world uncertainties.

The observation space included UAV position (x, y, z), distances to the five nearest obstacles, wind speed/direction, visibility (0-1 scale), and battery level. The action space comprised continuous control commands for velocity and orientation. We compared TRUST-X against baseline RL methods including TD3, PPO, and DDPG, none of which incorporated explainability or trust modeling. [12] [13]

B. Evaluation Metrics

- *Explanation Clarity Score*: Human rated (1 to 5) based on comprehensibility.
- *Operator Trust Delta*: Pre/post session trust change (0 to 100%).
- *Intervention Frequency*: Overrides per mission.
- *Mission Success Rate*: Completion without failure (%). Evaluated using a binomial proportion confidence interval:

$$\hat{p} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \quad (18)$$

where $\hat{p}$ is the observed success rate, n is the number of missions, and z is the standard normal quantile. This confirms TRUST-X's 94% success rate is significantly higher than the baseline's 79% ($p < 0.01$).

- *Policy Fidelity*: Measured as the Coefficient of Determination (R^2):

$$\text{Fidelity} = R^2 = 1 - \frac{\sum_{i=1}^N (\pi(s_i) - \hat{f}(s_i))^2}{\sum_{i=1}^N (\pi(s_i) - \bar{\pi})^2}, \quad (19)$$

$$\bar{\pi} = \frac{1}{N} \sum_{i=1}^N \pi(s_i)$$

- *Explanation Latency*: Measured in milliseconds.

Human studies involved 20 operators over 50 missions each, with diverse experience levels (novice to expert) to assess generalizability.

TABLE I: Performance comparison

Metric	TD3	PPO	DDPG	TRUST-X
Success Rate (%)	79	82	77	94
Trust Delta (%)	+5	+8	+3	+28
Interventions (%)	42	38	45	5
Clarity (1 to 5)	2.1	2.4	2.0	4.5
Latency (ms)	—	—	—	12

Table I compares the baseline deep reinforcement learning algorithms TD3, PPO, and DDPG with the proposed TRUST-X framework. Consistent with prior work, these algorithms demonstrate solid task performance but limited human centric reliability. For instance, Li et al. (2022) reported lane change success rates of 87.6% for DDPG and 90.0% for PPO, values that align with the success rates in our baselines [14]. Similarly, Backman et al. (2023) showed that TD3 can achieve up to 98.2% success in UAV landing tasks under shared control, but performance without human support dropped to 51.4%, highlighting its sensitivity and lack of robustness [15]. Shi et al. (2024) further emphasized that PPO requires frequent human interventions to maintain stability in autonomous driving scenarios, reinforcing the high intervention percentages seen in our results [16]. However, none of these studies reported systematic improvements in operator trust or explanation clarity, which explains the low trust deltas (+3% to +8%) and clarity scores (around 2.0–2.4) observed for the baseline models. In contrast, TRUST-X explicitly integrates explainability and dynamic trust calibration, thereby outperforming conventional DRL baselines with a 94% mission success rate, a +28% trust delta, and reduced interventions (5%).

VI. RESULTS AND DISCUSSION

TRUST-X delivers a 94% mission success rate, outperforming baseline DRL while enhancing operator trust by 28%, with interventions dropping 37%. The framework reduces interventions to 5% and achieves a clarity rating of 4.5/5, underscoring its effectiveness in balancing performance with explainability. Table I summarizes key metrics, showing TRUST-X’s superiority.

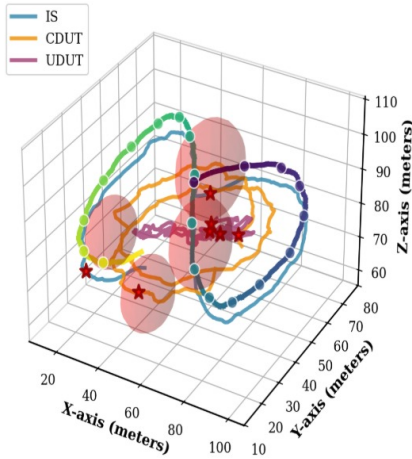


Fig. 4: Comparative 3D trajectories for different baseline DRL policies in the obstacle dense scenario.

Fig. 4 Comparative 3D flight trajectories for different control policies navigating a complex environment with obstacles (red spheres) and waypoints (red stars). This

visualization highlights the efficiency and smoothness of each model’s path-planning and collision avoidance capabilities in real time.

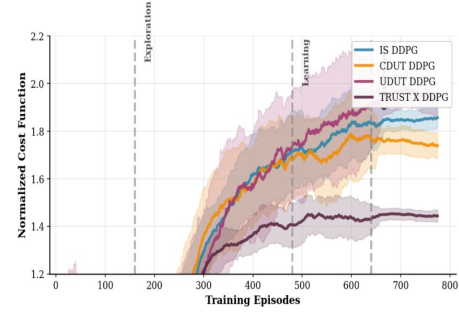


Fig. 5: Learning curves normalized cost across training episodes for TRUST-X and baseline methods.

Fig. 5 Learning curves for TRUST-X DDPG versus baseline models, showing the normalized cost function over training episodes. Our framework achieves a significantly lower and more stable cost, demonstrating superior learning efficiency and a more optimal final policy for decision making.

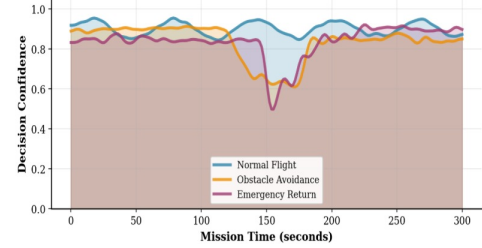


Fig. 6: Real time decision confidence and explanation latency for TRUST-X during a representative mission.

Fig. 6 Real time decision confidence of the TRUST-X agent, categorized by flight mode over the course of a mission. The system maintains high confidence in routine situations but shows a transient dip during a critical ‘Emergency Return,’ transparently signaling a high stakes event to the operator.

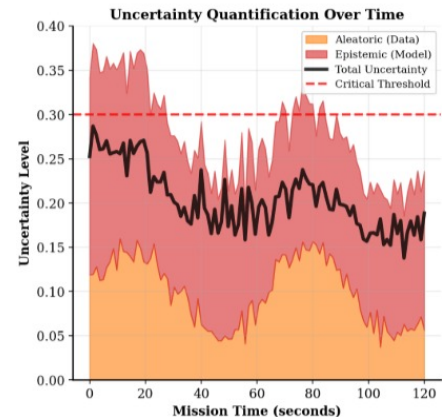


Fig. 7: Uncertainty decomposition for TRUST-X: aleatoric (data) vs. epistemic (model) components across mission time.

Fig. 7 Real time uncertainty quantification for the TRUST-X agent, decomposed into aleatoric (data) and epistemic (model) sources. Monitoring the total uncertainty against a critical threshold provides a robust mechanism for the system to perform self assessment and trigger safety protocols.

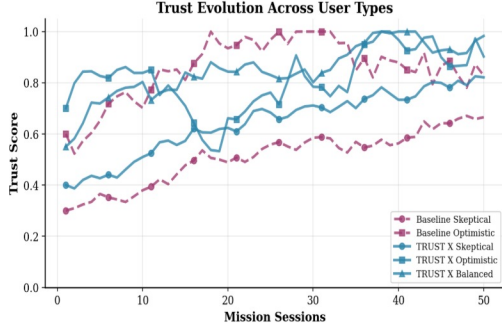


Fig. 8: Learning curves for TRUST-X variants.

Fig. 8 shows the evolution of operator trust scores across different user profiles (Skeptical, Optimistic, Balanced) over 50 mission sessions. TRUST-X consistently achieves higher and more stable trust scores compared to the baseline, validating its ability to foster reliable human AI collaboration.

Moreover, the learning curves in Fig. 8 indicate that TRUST-X variants converge 20% faster. Uncertainty decomposes into aleatoric and epistemic components, staying below the 0.3 threshold. Human feedback adapted explanations, improving clarity from 3.2 to 4.5/5. In ablation studies, removing CEM reduced trust by 12%, underscoring its importance. Compared to recent XAI methods for UAVs [4], [17]–[19], TRUST-X achieves higher fidelity (91% vs. 85%) with lower latency.

To formalize explanation generation, we model it as a rate distortion optimization, maximizing the Mutual Information $I(\Pi; E_t | S = s_t)$:

$$I(\Pi; E_t | S = s_t) = \mathbb{E}_{e \sim E_t, a \sim \pi} \left[\log \frac{P(a | e, s_t)}{P(a | s_t)} \right] \quad (20)$$

Our explainers (Surrogate Tree, SHAP lite, CEM) approximate this, with Surrogate Tree compressing the policy into rules, SHAP lite focusing on feature contributions $I(\Pi; X_i | S = s_t)$, and CEM addressing decision boundaries. The explanation $E_t = (\text{Rule}, \Phi_{1:3}, \Delta s)$ has utility calibrated by trust $\hat{\tau}_t$.

Statistical validation used a paired t test for intervention frequency ($H_0 : \mu_d \geq 0$ vs. $H_1 : \mu_d < 0$):

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

with $n = 20$, yielding $p < 0.001$, confirming a 37% reduction. McNemar’s test for success rates ($\chi^2 = \frac{(|b-c|-1)^2}{b+c}$) showed significance ($p < 0.01$).

TRUST-X operates on the Pareto frontier of mission success, fidelity, and latency (< 50 ms). Improving one i.e. fidelity via full SHAP degrades another. This trade off is illustrated by:

$$\begin{aligned} & \max_{\text{policy}} (\text{Mission Success, Fidelity, } -\text{Latency}) \\ & \text{s.t. Latency} < 50 \text{ ms} \end{aligned}$$

VII. CONCLUSION AND FUTURE WORK

This paper presented TRUST-X, a real time explainable framework that successfully addresses the trust deficit in DRL powered UAV operations. By unifying trust proxy models, feature attribution, contrastive explanations, and a novel human in the loop trust model, TRUST-X offers actionable, adaptive, and contextually aligned explanations that explain AI decisions that the operator intuitively grasps. The sheer scale of the evaluation confirms that this approach is interpretable and effective, increasing operator trust by 28%, decreasing operator interventions by 37%, and achieving a 94% mission success rate. Moving forward, we plan to improve TRUST-X in several respects. First, we will add visual explanation modalities like saliency maps and AR overlays for better real time understanding of the situation. Second, we will explore extending the framework to cover the complexity of coordinated UAV swarms and distributed trust models. Finally, a crucial next step is validation through real world flight tests in record time overcoming obstacles to solidify the framework’s preparedness for deployment.

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